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Characterization and microbiological evaluation of chitosan-alginate microspheres for cefixime vaginal administration

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ABSTRACT

Chitosan-alginate microspheres (MS) were developed for cefixime vaginal administration, to overcome problems associated with its oral administration. The effect of increasing drug-loading amount, by keeping the chitosan-alginate content constant, was investigated. Mucoadhesion studies indicated that all formulations assured *in situ* permanence longer than 2 h. Entrapment efficiency increased with drug loading concentration in the starting solution, reaching a plateau at 30 mg/mL indicative of the achievement of an optimal drug-to-polymer ratio. MS swelling properties increased with the entrapped drug amount, and, interestingly, water-uptake reached its maximum value at the same drug loading concentration of 30 mg/mL. The relationship found between MS water-uptake and drug release rate confirmed MS prepared with 30 mg/mL cefixime as the best formulation. Microbiological studies showed a relation between cefixime release rate from MS and *Escherichia coli* viability reduction, definitely indicating the selected MS formulation as the best for an effective local treatment of urogenital infections.

1. Introduction

Urogenital infections present significant problems for women's health. It is estimated that 1 billion women around the world suffer from urogenital infections, including bacterial vaginosis, urinary tract infections (UTIs) and yeast vaginitis. There is a clear link between vaginal microflora and UTIs (Sumati & Saritha 2009). *Escherichia coli* is one of the most common organisms found in the genital tract of non-pregnant (9–28%) and pregnant women (24–31%). It is considered to be a reservoir for vaginal and/or endocervical colonization in pregnant women and an important step in the development of urinary tract, intra-amniotic and puerperal infections through 'fecal-vaginal-urinary/neonatal' transmission (Sáez-López et al., 2016). Furthermore, in postmenopausal women, gradual depletion of estrogen supply leads to conversion of the vaginal flora from *Lactobacillus* to *E. coli* and other *Enterobacteriaceae*, thus increasing the incidence of recurrent UTIs (Khunda & Elneil, 2012). Therefore, eradication of *E. coli* from vaginal microflora is of therapeutic relevance, especially taking into account that UTIs are one of the most frequent clinical bacterial infections in women, accounting for nearly 25% of all infections (Pickard et al., 2016). Among the different kinds of UTIs, cystitis is the most common disease, particularly in young women, and *E. coli* is the causative microrganism in nearly 85% of cases (Schaeffer, Schaeffer, & Schaeffer, 2007). The effective treatment of acute UTIs still remains one of the most daunting challenges in clinical practice, due to high propensity to frequent recurrence, high prevalence rate, and associated co-morbidities that need to be addressed as well.

Cefixime (CFX), is a third generation cephalosporin antibiotic highly active against a broad range of Gram-negative and some Gram-positive aerobic bacteria, clinically used in the treatment of various susceptible infections, including respiratory tract infections, gonorrhea and UTIs (Martindale, 2012; Woodward & Fisher, 1999). CFX is very poorly soluble in water and, after oral administration, it is slowly and incompletely absorbed from the gastrointestinal tract, resulting in poor and variable bioavailability (40–50%) (Yash, Manoj, & Bhupinder, 2011). Moreover, CFX oral therapy can induce several collateral effects such as diarrhea, loose or frequent stools, abdominal or stomach pain, dizziness, headache, rash, nausea and vomiting (Naber, Wullt, & Wagenlehner, 2011).

Vaginal administration of CFX represents an interesting alternative

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to the oral route for the treatment of urogenital infections. In fact, local treatment of vaginal problems is becoming favored over oral approaches because of the achievement of higher local drug concentrations and the absence of drug interactions and interferences with the gastrointestinal tract (Vermani & Garg, 2000). However, despite their potential therapeutic usefulness, no CFX vaginal formulations are currently present on the market, and, at the best of our knowledge, no previous studies in this regard have been reported in literature.

Conventional vaginal formulations (including solutions, creams, foams, pessaries) have often a limited therapeutic efficacy and require a frequent dosing regimen, owing to the self-cleansing action of the vagina, namely the secretion of mucus and the humid site of administration. Bioadhesive vaginal delivery systems have been proposed to overcome such drawbacks, since they may adhere to the mucosa and extend the *in situ* residence time of the drug (Palmeira-de-Oliveira, Palmeira-de-Oliveira, & Martinez-de-Oliveira, 2015). Carbohydrate mucoadhesive polymers proved to play a basic role in improving the performance of vaginal delivery systems, related to their ability to prolong the local retention of the dosage form and assure a sustained *in situ* drug delivery (Acartürk, 2009; Bassi & Kaur, 2012; Valenta, 2005). Among such polymers, chitosan (a N-deacetylated product of chitin) and alginate (a natural anionic polysaccharide based consisting of beta-D-mannuronic acid (M) and alpha-L-guluronic acid (G) repeating units) appear of particular interest, in virtue of their favorable biopharmaceutical properties, such as excellent tolerability, biocompatibility, non-toxicity, biodegradability. Both chitosan and alginate polymers have been successfully used as excipients for a variety of drug delivery systems, including vaginal formulations (Bonferoni et al., 2006; Duttagupta, Jadhav, & Kadam, 2015; Perioli et al., 2008; Tønnesen & Karlsen, 2002; Valenta, 2005). Moreover, the combined use of such polymers can be more effective than alginate or chitosan alone in limiting the release of the entrapped materials, in virtue of the formation of interactions between alginate carboxyl groups and chitosan amine groups (Gazori et al., 2009; Pasparakis & Bouropoulos, 2006), and it has been successfully exploited also in vaginal delivery (Abruzzo et al., 2013; Casella, Maestrelli, Scuota, Bazzucchi, & Mura, 2015; El-Kamel, Sokar, Naggar, & Al Gamal, 2015; Maestrelli, Cirri, Mennini, & Mura, 2014). Moreover a possible antibacterial action of chitosan has been reported (Goy, Morais, & Assis, 2016), which could be particularly useful in the present case, giving a synergic effect with CFX.

Mucoadhesive multiparticulate systems can offer several advantages for vaginal administration compared to single dosage forms such as vaginal tablets (El-Kamel et al., 2015) or vaginal inserts (Abruzzo et al., 2013), since they can cover a wider mucosa area and allow a prolonged drug permanence on the vaginal mucosa surface and a controlled release over time (Albertini et al., 2009; Mehta et al., 2012; Martín-Villena et al., 2013).

Based on these premises, the aim of this work was the development of a new vaginal formulation based on chitosan-coated calcium alginate microspheres, which should allow a more effective treatment of local bacterial infections susceptible to this drug, overcoming the low bioavailability and side effects problems related to CFX oral administration. Microspheres were prepared by ionotropic gelation technique, and the effect on their performance of loading with increasing CFX amounts, by keeping the chitosan-alginate content constant, was investigated. Each batch was characterized in terms of mean weight and diameter, morphology (by ESEM and optical microscopy), entrapment efficiency and loading capacity. *In vitro* water-uptake, mucoadhesion, drug release and microbiological studies were carried out in order to investigate the ability of the microspheres to adhere to the vaginal mucosa and suitably release the drug over time and to evaluate their actual antimicrobial activity toward *E. coli*.

2. Materials and methods

2.1. Materials

Cefixime (CFX) was kindly donated by Menarini Group (Italy). Sodium alginate (mannuronic acid/guluronic acid ratio ≈1.56; molecular weigh 80–120 kDa), chitosan (CS) (deacetylation degree 75–85%; mean molecular weight 50 kDa,) and calcium chloride, were purchased from Sigma (Saint Louis, MO, USA). Porcine vaginal mucosa coming from slaughter pigs was kindly provided by Doctor De Angelis (Local Health Authority of Arezzo, Italy). All other chemicals and solvents were of analytical reagent grade.

2.2. Preparation of microspheres (MS)

CFX-loaded MS were prepared by ionotropic gelation technique, opportunely modifying the methods described by Mennini, Furlanetto, Cirri, and Mura (2012) and by Maestrelli, Zerrouk, Cirri, and Mura (2015), using a two-step or one-step procedure for CS coating. In the two-step method, 10 mL of a 3% w/v sodium alginate solution, containing increasing drug concentrations (0–40 mg/mL), was dropped (needle inner diameter 0.7 mm, flow rate 1 mL/min), under magnetic stirring, in a 10% w/v calcium chloride aqueous solution, used as cross-linker, and allowed to stand for 15 min from the dropping stop. The MS were then filtered, washed with distilled water, placed in the coating solution containing 1.5% w/v CS in CH_3COOH 5% v/v and allowed to stand for 15 min. In the one-step method the 3% w/v sodium alginate solution was dropped in the aqueous solution containing both the cross-linker (10% w/v calcium chloride) and the coating agent (1.5% w/v CS in CH_3COOH 5% v/v) and allowed to stand for 30 min. The MS were then collected, washed with distilled water and filtered. All the batches were dried in an oven at 37° C for 24 h.

2.3. Characterization of microspheres (MS)

2.3.1. Determination of mean diameter and weight

The mean diameter of MS was calculated on 20 units for each batch and measured with a stainless steel digital caliber, as the mean of three different measures performed on each particle. The mean weight of polymeric MS was calculated on 20 units from each batch and measured using an analytical balance (Mettler Toledo M3, Switzerland).

2.3.2. Morphological studies

Shape and general appearance of MS were investigated by optical microscopy (Hund Medicus AFL microscope, magnification 10×). Photographs of each freshly prepared batch were taken with a Canon Power Shot A60 digital camera with 1600 × 1200 pixel magnification. The surface morphology of MS was examined in detail using a FEI Quanta 200™ environmental scanning electron microscope (ESEM), with a spatial resolution at 30 kV of 3 nm (FEI Company, Oregon, USA) equipped of EDS-X-ray Microanalysis System (EDAX, software EDAX Genesis).

2.3.3. Drug entrapment efficiency and loading capacity of MS

Drug entrapment efficiency (EE%) was calculated by an indirect method (González-Rodríguez, Holgado, Sánchez-Lafuente, Rabasco, & Fini, 2002; Maestrelli et al., 2015) according to the following equation:

$$EE\% = ((Qt - Qr)/Qt) \times 100$$

where Qt is the drug content initially added during MS preparation and Qr the sum of the drug content recovered in the aqueous solutions after separating and washing the MS.

CFX was assayed spectrometrically at 288 nm (UV–vis 1600 Shimadzu spectrophotometer, Tokyo, Japan). The range of the calibration curve was from 0.43 to 2.18 mg L^{-1}. The obtained regression

equation was Y = 0.042 X − 0.029 (R^2 = 0.996). It has been verified that the other components of MS did not interfere with the drug assay. Moreover, UV spectra did not reveal the presence of degradation products of CFX (Gandhi & Rajput, 2009).

Loading capacity (LC%) was calculated according to the following equation:

LC% = (weight of entrapped drug/weight of MS). ×100

Each result is the mean of three separate determinations (C.V. < 4%).

2.3.4. Determination of the MS drug:polymers ratio

The drug:polymers ratio of each MS batch was calculated on the basis of the total weight of the obtained MS and the total weight of entrapped drug (determined by the drug entrapment efficiency, as described above).

2.3.5. Swelling properties of MS

MS swelling properties were evaluated according to the method described by Jug, Maestrelli, and Mura (2012), by measuring the uptake dynamics of a pH 5.5 phosphate buffer solution, used to simulate the vaginal medium. In fact, this is the estimated average value of vaginal pH in the presence of bacterial pathogens (Caillouette, Sharp, Zimmerman, & Roy, 1997). Briefly, a sponge (5 × 6.5c × 3 cm^3, SantexGmbH, Wald-Michelbach, Germany) was fully soaked in the pH 5.5 buffer solution and put in a Petri dish filled with the same medium to a 1 cm height, to maintain the sponge soaked during the experiment. Square filter papers (10 × 10 mm, Filtros Anoia SA, Barcelona, Spain), used as sample holders, were also soaked in the medium and put on the top of the sponge. The Petri dish was then covered and after 30 min equilibration at 37 °C, the wet filter papers were weighed (W_0). Accurately weighed MS (n = 5, W_1), were placed on the wet filter papers. At given time intervals the devices were removed from the medium and weighed (W_2). The swelling index (I), taken as a measure of water uptake, was calculated by the following equation:

$$I = \frac{W2 - W0}{W1} \times 100$$

Each test was performed in triplicate (C.V. < 4%).

2.3.6. Mucoadhesion studies

MS mucoadhesion properties were evaluated by measuring the ex vivo residence time using porcine vaginal mucosa as a model substrate, according to the methods of Liu et al. (2005) and Mennini, Casella, Cirri, Maestrelli, and Mura (2016) suitably modified. Briefly, excised porcine vaginal mucosa, obtained from a local slaughterhouse, was carefully separated from underlying tissue, washed with pH 5.5 phosphate buffer, cut into small pieces of 40 × 25 mm size and put vertically in a chamber thermostated at 37 °C. Then, 30 MS, previously swollen 20 min in pH 5.5 phosphate buffer, were placed over the tissue section with a light pressure and subjected to a continuous flux (9 mL/min) of the pH 5.5 phosphate buffer solution miming the vaginal fluid, to simulate and stress the physiological conditions. MS mucoadhesion ability was monitored for 2 h and expressed as time of mucoadhesion and as percent of MS remained on the mucosa at the end of the experiment (mean of three determinations).

2.3.7. In vitro drug release studies

In vitro drug release studies were carried out using a simulated dynamic vaginal system, based upon the maximum volume and secretion rate of mucus in the vagina under stress conditions, according to the method of Bassi and Kaur (2015), slightly modified. Freshly excised porcine vaginal tissue, obtained from a local slaughter house, was carefully separated from underlying tissue and cleaned, cut into portion of 40 × 25 mm and mounted inclined at 30° with the mucosal side facing upwards in a chamber thermostated at 37 °C. 30 MS, previously swollen 20 min in pH 5.5 phosphate buffer, were placed over the tissue section with a light pressure and subjected to a continuous flux (9 mL/min), obtained by a peristaltic pump, of the pH 5.5 phosphate buffer solution miming the vaginal fluid. Samples of the perfused buffer solution were collected at predetermined time intervals and spectrometrically assayed for drug content as described above. The results are the mean of three experiments.

2.4. Microbiological studies

2.4.1. Antimicrobial susceptibility testing (AST)

MIC (minimal inhibitory concentration) of CFX was determined by the serial two-fold micro-dilution method in Mueller-Hinton broth (Merck, Germany), according to the European Committee on Antimicrobial Susceptibility Testing (EUCAST) Discussion Document E. Dis 5.1 (EUCAST, 2003) using 96-well flat-bottom microtiter plates (Kartell, Italy). *Escherichia coli* ATCC 10538 was used as the model strain. Inocula were prepared by dispersing a fresh culture of *E. coli*, cultured aerobically on tryptic-soy agar plates (Merck, Germany) during 18 h at 37 °C, in pH 7.4 phosphate buffer (PBS) adjusted up to 0.5 McFarland standard units (1.5×10^8 CFU mL^{-1}), adjusted using a nephelometer (Densimat, BioMerioux, France). The CFX concentration ranged from 12.5 up to 0.006 μg mL^{-1}. Microdilution plates were incubated aerobically at 37 °C for 18 h. After that, the viability of *E. coli* was determined using 2,3,5-triphenyl-tetrazolium chloride assay (TTC, Sigma-Aldrich, USA), following a previously described procedure (Klančnik, Piskernik, Jeršek, & Možina, 2010). Briefly, 50 μL of TTC solution (50 mg/mL in PBS) was added to each well of plates cultured with *E. coli*, followed by 3 h incubation at 37 °C and subsequent addition of isopropyl alcohol containing 4% HCl (v/v) to dissolve the formazan crystals formed. Then the absorbance was measured at 560 nm using iEMS MF Microplate reader (Themo Lab Systems, USA). MIC values were calculated as the lowest drug concentration causing more than 80% reduction in cell viability when compared to the control (untreated bacterial cells). All experiments were done in triplicate.

2.4.2. Time-kill assay

The time-kill assay was performed in Mueller-Hinton broth using *E. coli* as the test strain. Pure CFX or as MS formulation was added directly in the liquid nutrient medium inoculated with *E. coli* at a density of 1.5×10^8 CFU/mL. In all cases, the CFX concentration was set to 0.27 μg/mL, that corresponds to 3 times the MIC value determined according to the protocol described in Section 2.4.1. The culture flasks were incubated aerobically at 35 °C in an orbital shaker at 170 revolutions/min. After 0, 1, 3, 6, and 9 h of incubation, viable colony counts were performed by taking 100 μL of culture and making 10-fold serial dilutions in PBS followed by spread-plating of each sample on tryptic-soy agar (Merck, Germany). Plates were incubated for 18 h at 37 °C and then last two dilutions containing growing colonies of *E. coli* were counted as number of living cells for each time point.

Furthermore, the changes in metabolic activity of *E. coli* at the same time points were also monitored, by a modified TTC assay, using the TTC solution in concentration of 0.2 mg/mL. All other experimental conditions were as described above. The metabolic viability (%) of *E. coli* at each time point was calculated according to the following equation:

$$viability = \frac{A_{560\ nm\ of\ the\ sample}}{A_{560\ nm\ of\ the\ CFX-free\ control}} \times 100\%$$

Each time–kill experiment was performed in triplicate.

2.5. Statistical analysis

All data were statistically analyzed by ANOVA (one-way analysis of variance) followed by the Student-Neuwman Keuls multiple

comparison post-test (GraphPad Prism, version 6, San Diego, CA, USA). The differences were considered statistically significant when $P < 0.05$.

3. Results and discussion

The present study was aimed to develop mucoadhesive CS-coated Ca alginate MS for CFX vaginal delivery, exploiting the favourable properties of both polymers to provide an *in situ* prolonged residence time and extended drug release.

The first phase of the study was devoted to set the CS coating method. MS were then prepared according to the ionotropic gelation method, following a one-step or two-step procedure, using a fixed CFX loading concentration in the starting solution of 5.0 mg/mL. According to the two-step method, the CS coating was performed after Ca alginate MS obtainment, while with the one-step method MS formation and coating occurred almost simultaneously. MS of homogeneous and similar appearance in terms of shape, size (mean diameter 1.41 *vs* 1.39 mm) and weight (mean weight 2.9 *vs* 3.1 mg) were obtained with both methods. No significant differences ($p < 0.05$) were observed in drug entrapment efficiency, which resulted 58.15 ± 2.07% in the first case and 60.32 ± 1.83% in the second one. Based on these findings, the simpler one-step process was selected.

A series of MS was then prepared, by keeping the chitosan-alginate content constant (in a 1:2 w/w ratio, corresponding to a 1:1 molar ratio) and progressively increasing the drug loading concentration in the starting solution from 5.0 up to 40 mg/mL. The resulting MS batches were characterized in terms of mean diameter, weight, entrapment efficiency, loading capacity and drug to polymers w/w ratio, as reported in Table 1.

The results showed that the mean diameter of the different MS batches resulted homogeneous (as proved by the a low standard deviation around the mean value) and gradually increased with increasing the loaded drug amount passing from 1.39 mm for empty MS (MS_0) up to 1.70 mm for MS_4 and MS_5; in parallel, their mean weight progressively increased, passing from 2.14 mg for MS_0 up to about 4.4 mg for MS_4 and MS_5.

In the same way, EE% increased with increasing the drug concentration in the starting solution up to 30 mg/mL (MS_4), while it did not further increase when the drug concentration rose to 40 mg/mL (MS_5). This finding could be explained by assuming that the drug-to-polymer ratio achieved in the MS_4 formulation (around 1:1 w/w) was the optimal one to maximize the drug entrapment and the polymeric matrix was not able to effectively retain higher drug amounts.

MS shape, general appearance and surface morphology were examined by optical end electronic microscopy. Photographs of each batch, obtained with a 10× magnification are reported in Fig. 1. As shown, MS exhibited a different color, varying from light yellow up to brown; this effect was found to be related to the increasing amount of drug entrapped. All batches presented a homogeneous aspect, exhibiting a relatively regular and almost round shape with a smooth surface.

These results were confirmed by ESEM analysis (Fig. 2), where the surface of MS appeared quite smooth and uniform for both unloaded (A) and CFX-loaded (B) MS, and their nearly spherical form was evident. In Fig. 2C, a microphotograph of a section of loaded MS is reported, evidencing the presence of crystals in the internal core, attributed to the drug presence.

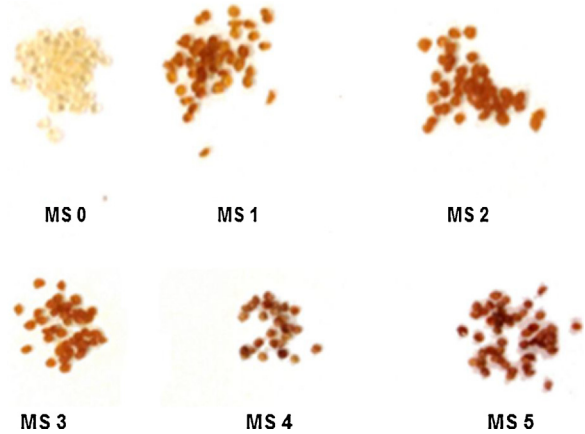

Fig. 1. Photographs of chitosan-alginate microspheres unloaded (MS_0) or loaded with increasing amounts of cefixime (MS_1 − MS_5) (see Table 1 for MS composition).

The Energy-Dispersive-X-ray analysis performed on this zone (Fig. 2D), evidenced the presence of sulfur, component of CFX molecule, thus confirming the presence of the drug, which appeared prevalently distributed within the polymeric matrix and not on the MS surface.

The swelling behavior of the different MS batches in a pH 5.5 buffer solution simulating the vaginal medium was evaluated and the results reported in Fig. 3. As shown, MS_4 displayed the highest water uptake ability, immediately followed by MS_5, and then by MS_3, MS_2, MS_1 and finally by empty MS (MS_0). A possible explanation of such finding could be the progressive loss of the polymeric matrix structure when passing from empty MS_0 up to MS_4 and MS_5, due to the increasing amount of entrapped drug, and then the concomitant increase in the drug to polymers w/w ratio, with the formation of pores and/or channels in the polymeric matrix, which facilitated the water uptake.

In particular, the low water-uptake shown by the matrix in the absence of drug (MS_0) can be ascribed to the strong electrostatic interactions between the polymers (which are present in an almost 1:1 molar ratio), being most of CS amino groups and alginate carboxylic groups charged at pH 5.5 (chitosan pKb ≈ 6.3 and alginate pKa ≈ 3.5) (Abruzzo et al., 2013). In the presence of CFX, due to its acidic character (pKa ≈ 2.0–2.5), it competed with alginate for interacting with the CS amino groups, thus allowing more charged alginate carboxylic groups to freely interact with water molecules, with a consequent increase of water-uptake ability. This phenomenon, similarly to what found for drug entrapment efficiency, reached a maximum when CFX concentration in the starting solution was 30 mg/mL (MS_4), confirming the hypothesis of the achievement of an optimal drug-to-polymer ratio in the matrix of this MS batch.

As for the release behavior (Fig. 4), in all cases a biphasic profile was observed, represented by an initial "burst" effect, during the first

Table 1
Mean diameter, weight, drug entrapment efficiency (EE%), loading capacity (LC%), chitosan (CS) content and drug/polymers ratio of microsphere (MS) batches loaded with increasing cefixime (CFX) amounts, keeping chitosan and alginate amounts constant.

Batch	c(CFX) in loading solution (mg/mL)	mean diameter (mm ± s.d)	mean weight (mg ± s.d)	EE%	LC%	CFX/polymers w/w ratio	CS content%
MS_0	0	1.39 ± 0.08	2.14 ± 0.12	0	0	–	33.3
MS_1	5	1.41 ± 0.04	3.11 ± 0.05	60.3 ± 1.8	22.6 ± 0.6	1:3.5	25.8
MS_2	10	1.53 ± 0.09	3.82 ± 0.09	67.3 ± 2.2	34.2 ± 1.0	1:2	21.9
MS_3	20	1.61 ± 0.02	4.13 ± 0.05	75.9 ± 4.1	39.0 ± 2.0	1:1.6	20.3
MS_4	30	1.70 ± 0.02	4.40 ± 0.06	84.8 ± 5.1	46.2 ± 2.6	1:1.1	17.9
MS_5	40	1.69 ± 0.02	4.41 ± 0.03	83.1 ± 3.6	47.4 ± 1.9	1:1.1	17.5

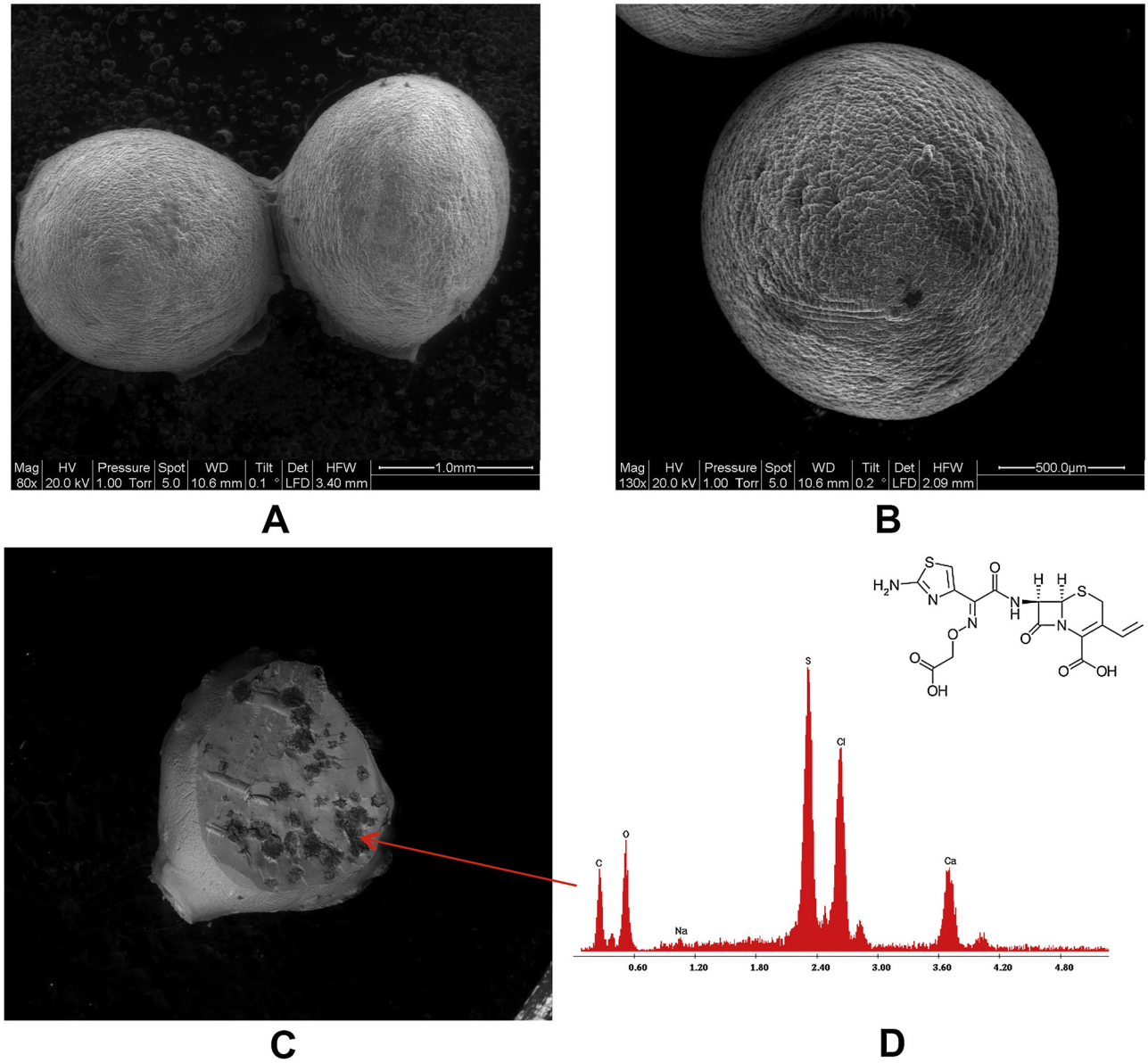

Fig. 2. ESEM images of unloaded (A) or cefixime-loaded chitosan-alginate microspheres (MS_3) (B) and of a section of loaded MS_3, with the related EDS X-ray fluorescence spectrum (see Table 1 for MS composition).

20–30 min, attributable to the dissolution of the drug portion adsorbed on the MS surface and to the drug rapid diffusion from the matrix layers nearest to the external surface; this phase was then followed by a prolonged, nearly linear, slow release phase, due to the diffusion of drug molecules through the internal MS core.

The drug release rate from the various MS appeared to be closely related to the MS swelling properties. In fact, MS_4, which exhibited the highest water uptake, also showed the highest drug release rate, probably in virtue of the greater amount of solvent penetrating in the polymeric matrix (Kılıçarslan & Baykara, 2003). The drug release mechanisms include the penetration of the medium inside the polymeric MS matrix, the swelling of the alginate network which favors the movement of drug molecules and their subsequent diffusion out of the CS coat (Pasparakis & Bouropoulos, 2006). Moreover, regardless of the different CS content, all MS batches did not show evidence of erosion phenomena at the end of the release test.

Mucoadhesion studies, performed on excised porcine vaginal mucosa under a continuous flux (9 mL/min) of the pH 5.5 phosphate buffer solution to mime the vaginal fluid in the presence of bacterial infections (Caillouette et al., 1997), and stress the physiological secretion rate, indicated that all CFX-loaded MS batches were endowed with good mucoadhesion properties. In fact, mucoadhesion times longer than two hours were recorded for all batches, and in all cases, 100% of tested MS remained adherent to the mucosa surface until the end of the experiment, irrespective of the CS content in the formulation, suggesting that all the proposed formulations should be able to provide a suitable in-situ residence time of the drug.

Microbiological studies were carried out using *E. coli* as a model strain, taking into account its pathogenic potential as well as the fact that it is a non-fastidious organism easy to cultivate. The MIC of CFX against *E. coli*, assessed by microdilution method was 0.09 µg mL^{-1}, which is in agreement with literature data (Andrews, 2001), thus confirming the validity of the selected model.

In order to verify the therapeutic efficiency of the developed MS formulations, the drug release data were then analysed in terms of MIC value. The concentration of CFX released at the first time point (5 min) ranged from a minimum of 3.5 mg/mL (MS_1) up to a maximum of 13.9 mg/mL (MS_4), that is from 39000 up to 155000 times higher than

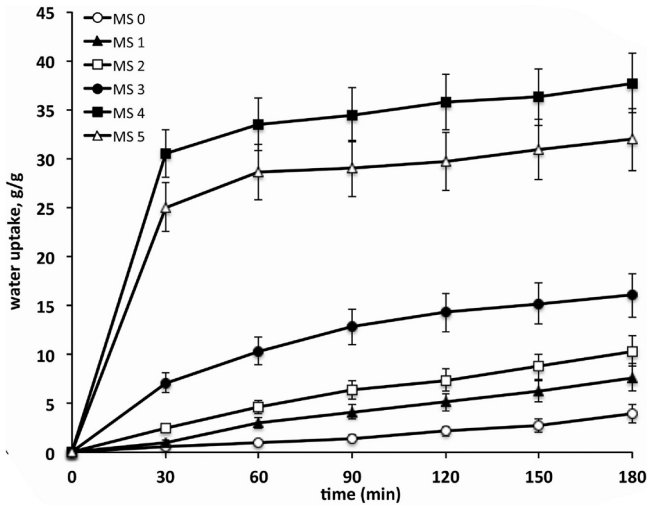

Fig. 3. Water uptake profiles over time of the different cefixime-loaded chitosan-alginate microspheres batches (see Table 1 for MS composition).

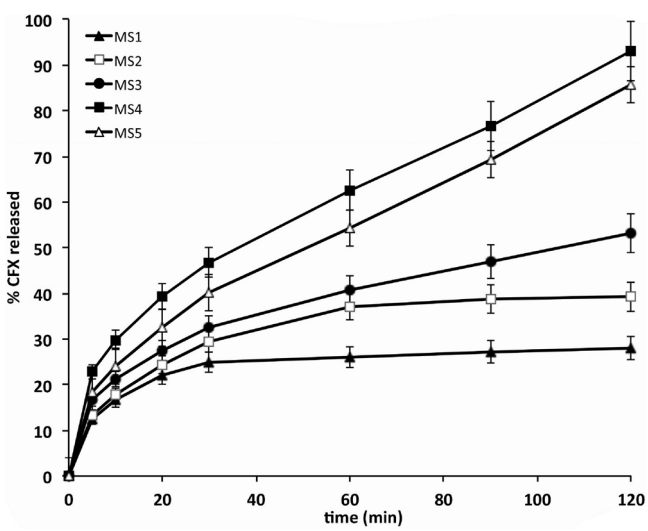

Fig. 4. Cefixime (CFX) release profiles from the different chitosan-alginate microspheres batches (see Table 1 for MS composition).

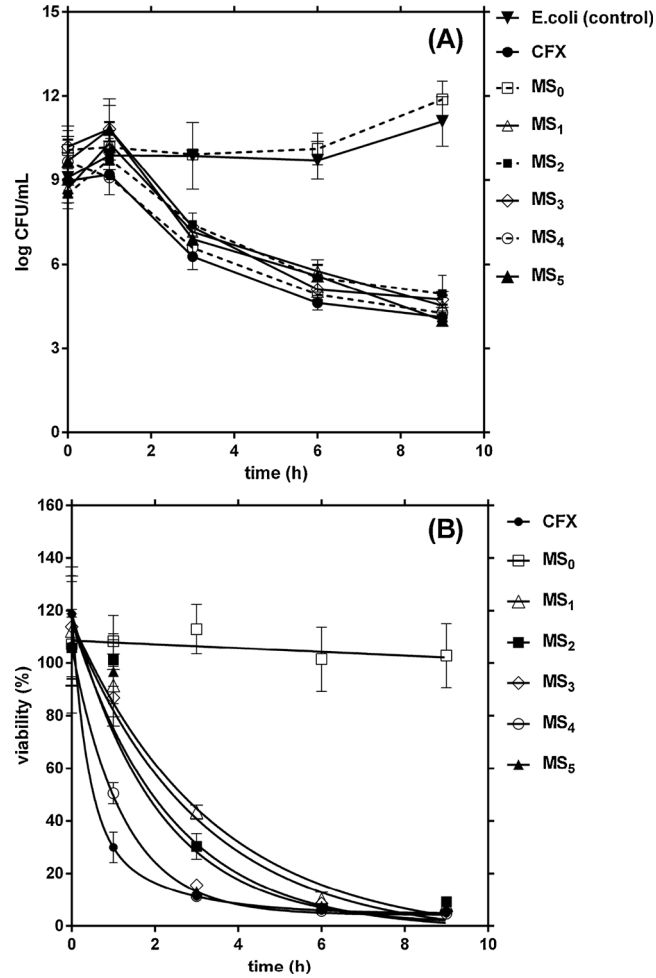

Fig. 5. Time-kill curves of *E. coli* treated with cefixime (CFX) alone or as chitosan-alginate microspheres formulations (see Table 1 for microspheres composition). The antimicrobial activity of the tested formulations was assessed by monitoring the changes in number of viable bacteria (A) and in bacterial metabolic activity (B) as a function of time.

MIC, and it further rise up during the release test (120 min). This clearly demonstrates the potential of developed MS formulations to release high CFX doses that could diffuse into the mucosal environment of vagina, providing local achievement of therapeutically relevant drug doses and efficient eradication of susceptible microbial strains.

To evaluate the effect of CFX microencapsulation on its antimicrobial efficacy and to determine the potential synergistic effect with the polymers used for MS preparation, a time dependent "time-kill" assay was performed. It is considered as the most appropriate method for determining the antimicrobial effect of a given formulation, providing valuable information about the interaction dynamic between the antimicrobial agent and the microbial strain. This is particularly important for formulations with limited residence time on the application site and could provide more relevant estimation of the clinical outcome achieved by the application of such formulation (Hoseinzadeh et al., 2017; Pappalardo et al., 2009). The antimicrobial activity of the tested formulations was assessed by monitoring the variation in the number of viable bacteria (Fig. 5A) and of the bacterial metabolic activity (Fig. 5B), both as a function of the time. In all cases the CFX concentration was set to 3 times the MIC value determined previously.

As for the reduction in viable bacteria number (Fig. 5A), the antimicrobial activity of all the drug-loaded MS formulations was comparable to that of pure CFX and there were no statistically significant differences between the antimicrobial activity presented by the drug and the MS_1-MS_5 formulations at all the given time points ($p > 0.05$). On the other hand, CFX-free microspheres (MS_0) had no effect on the number of *E. coli*, showing a trend comparable to that observed for the control sample, despite the reported bacteriostatic effect of CS against *E. coli* (Benhabiles et al., 2012). The lack of effect of CS was probably due to its low concentration in the formulation, which was evidently lower than its MIC value against *E. coli* (0.01%).

On the contrary, when monitoring the changes in *E. coli* metabolic activity (Fig. 5B), a statistically significant difference was observed between the different MS formulations at different times. For example, 1 h after treatment with MS_1, MS_2 and MS_3, the metabolic activity of *E. coli* was reduced only by approximately 10–20% in comparison to the control, while, after the same time, MS_4 decreased the metabolic activity of *E. coli* by more than 50%. After 3 h, the efficiency of MS_4 was comparable to that of MS_5 and MS_3 and still somewhat more pronounced than that of MS_1 and MS_2, while after 6 h, all formulations showed comparable efficiency.

Interestingly, the observed trend shown by the different MS batches in the reduction of the *E. coli* metabolic activity was analogous to that found for the drug release rate (Fig. 4). In fact, the slow and incomplete drug release form MS_1-MS_2 formulations reflected in a lower antimicrobial activity. Formulations MS_3 and MS_5 released the drug at lower rates than MS_4, therefore their efficiency was comparable only after longer time periods. On the other hand, MS_4 formulation, which

showed the highest drug release rate, also exhibited the highest antimicrobial effect. However, the decrease in *E. coli* metabolic activity observed in the presence of plain CFX after 1 h was significantly higher than that observed for MS_4 formulation, exceeding 70%. This finding can be explained by taking into account that in the case of MS formulation an initial lag time must be considered, due to the drug release from the polymeric matrix. In fact after 2 h, the antimicrobial activity of MS_4 formulation was comparable to that of the pure drug. Moreover, a longer *in situ* residence and a more prolonged effect is expected from CFX as MS formulation than as simple aqueous solution. Therefore, CFX-loaded MS_4 confirmed as the formulation of choice, able to assure a suitable in-situ residence time of the drug, and provide an efficient and prolonged drug release and an effective antimicrobial action.

4. Conclusion

A new vaginal formulation of CFX based on MS made of carbohydrate-mucoadhesive-polymers, has been developed, aimed to provide an effective treatment of urogenital infections and overcome the problems of poor bioavailability and frequency of side effects appearance related to the oral drug administration. Alginate and chitosan were selected as mucoadhesive polymers for MS preparation, in virtue of their well-known favorable physicochemical and biopharmaceutical properties, particularly suitable for vaginal formulations. The influence on the MS performance of loading with increasing CFX concentrations was carefully investigated.

All MS batches prepared by keeping the chitosan-alginate content constant and increasing drug loading amounts ($MS_1 - MS_5$) were able to assure a prolonged *in situ* permanence, longer than 2 h, on excised porcine vaginal mucosa, showing good mucoadhesion properties. Entrapment efficiency increased with increasing the starting loading drug concentration up to 30 mg/mL (MS_4 formulation), and then showed a plateau, indicating that an optimal drug-to-polymer ratio has been reached at such drug concentration. Swelling studies confirmed this finding, showing a progressive increase of water uptake ability of MS formulations, up to a maximum for MS_4 formulation. Moreover, a direct relation was found between MS water uptake and drug release rate, confirming MS_4 formulation as the most effective one.

Microbiological studies definitely supported these results, pointing out a direct relation between *in vitro* CFX release rate from MS and% reduction of metabolic activity of *E. coli*, used as model strain, and proved that the antibacterial activity of CFX as MS_4 formulation was comparable to that of the pure drug.

The developed CFX MS_4 formulation could be administered inside a capsule, by a commercially available vaginal applicator, for an effective treatment of local vaginal and urogenital diseases. After dissolution of the capsule, a distribution of MS over the entire vaginal mucosa is expected, as observed for other multiparticulate vaginal drug delivery systems (Mehta et al., 2012), and their good mucoadhesion power and extended-release properties would assure a prolonged *in situ* activity of CFX.

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